

20. Pipelining (5)

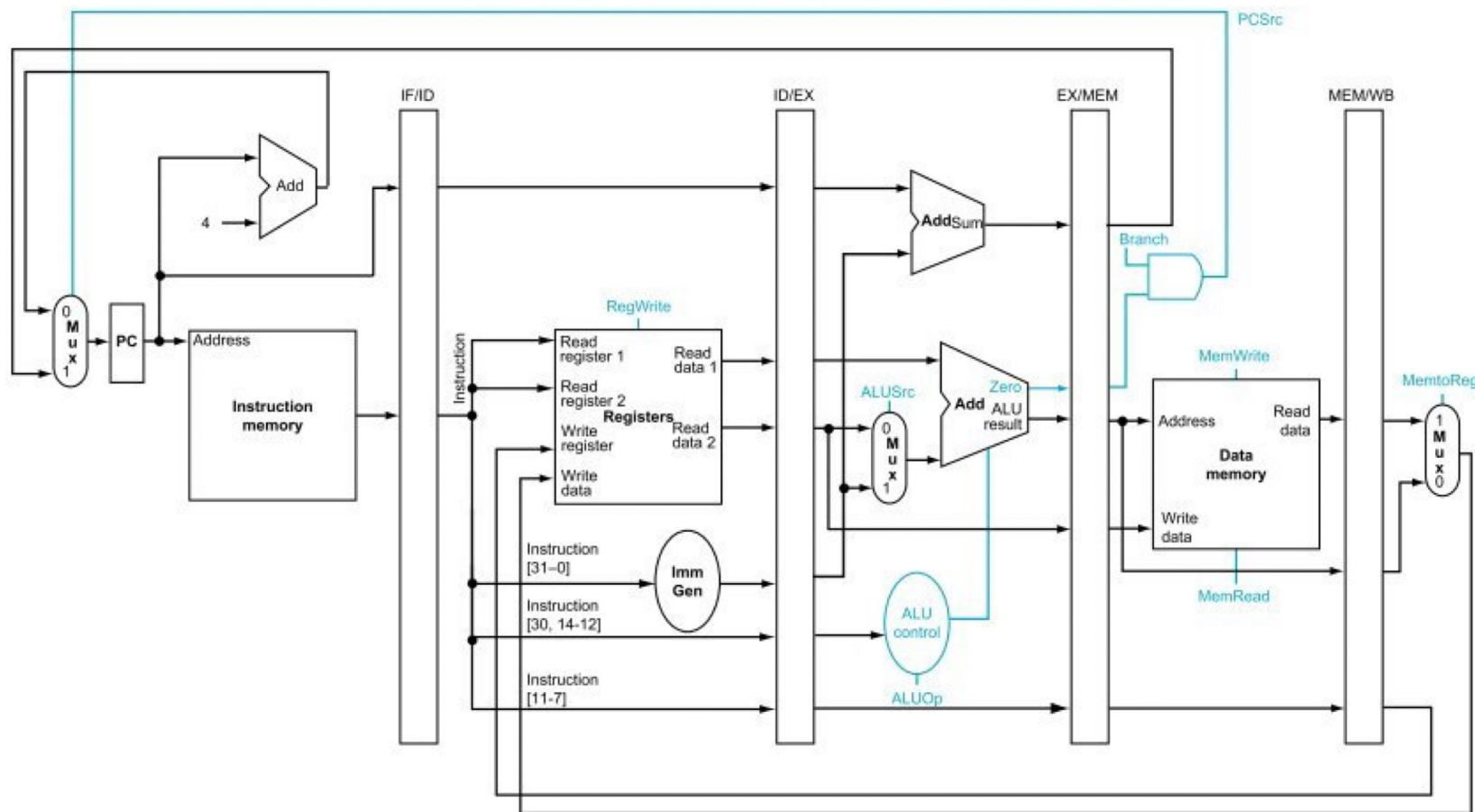
Example: Decoder

```
unique case ({instr_alu[31:25], instr_alu[14:12]})
  // RV32I ALU operations
  {7'b000_0000, 3'b000}: alu_operator_o = ALU_ADD; // Add
  {7'b010_0000, 3'b000}: alu_operator_o = ALU_SUB; // Sub
  {7'b000_0000, 3'b010}: alu_operator_o = ALU_SLT; // Set Lower Than
  {7'b000_0000, 3'b011}: alu_operator_o = ALU_SLTU; // SLT Unsigned
  {7'b000_0000, 3'b100}: alu_operator_o = ALU_XOR; // Xor
  {7'b000_0000, 3'b110}: alu_operator_o = ALU_OR; // Or
  {7'b000_0000, 3'b111}: alu_operator_o = ALU_AND; // And
  {7'b000_0000, 3'b001}: alu_operator_o = ALU_SLL; // Shift Left Log
  {7'b000_0000, 3'b101}: alu_operator_o = ALU_SRL; // Shift Right Log
  {7'b010_0000, 3'b101}: alu_operator_o = ALU_SRA; // Shift Right Arith
```

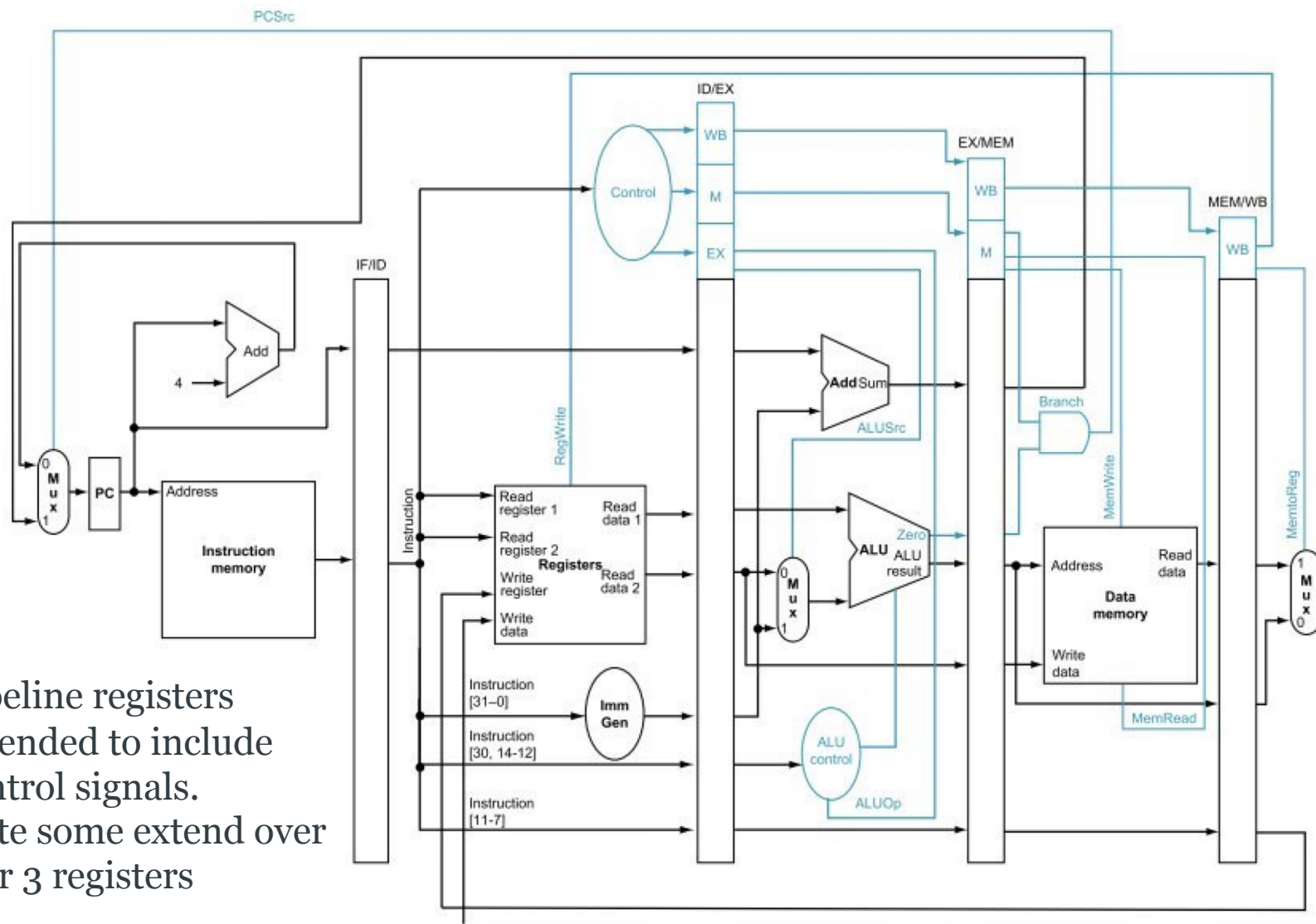
- R-type instruction (OPCODE_OP = 7'h33)
 - concatenates the funct7 and funct3 fields
 - alu_operator_o is of type alu_op_e (see earlier)
 - bit pattern is undefined (enumerated type)

Status Signals in Pipeline

- Zero flag used for branching – not pipelined

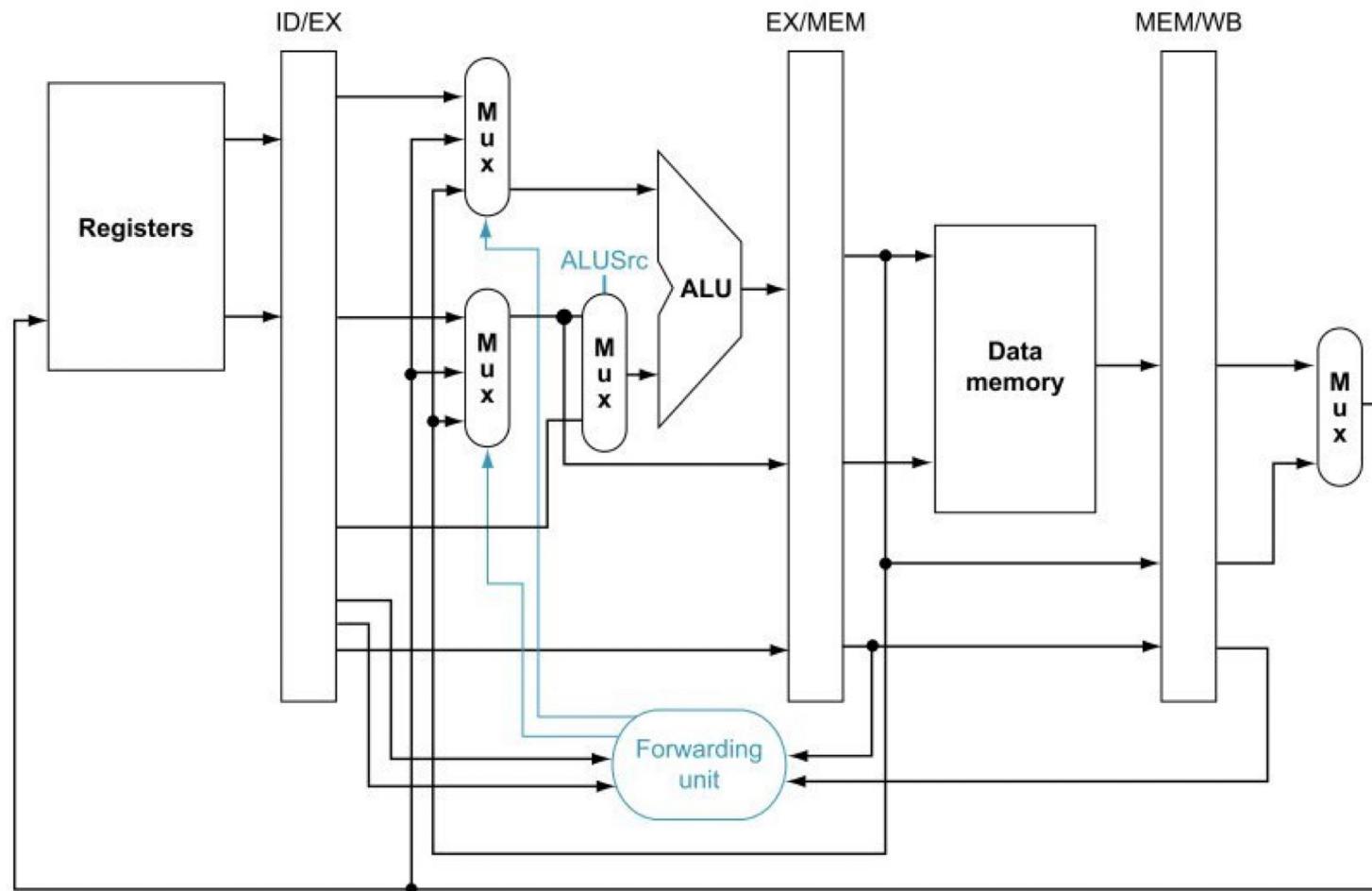


Control Signals are pipelined



Pipeline registers
extended to include
control signals.
Note some extend over
2 or 3 registers

Forwarding



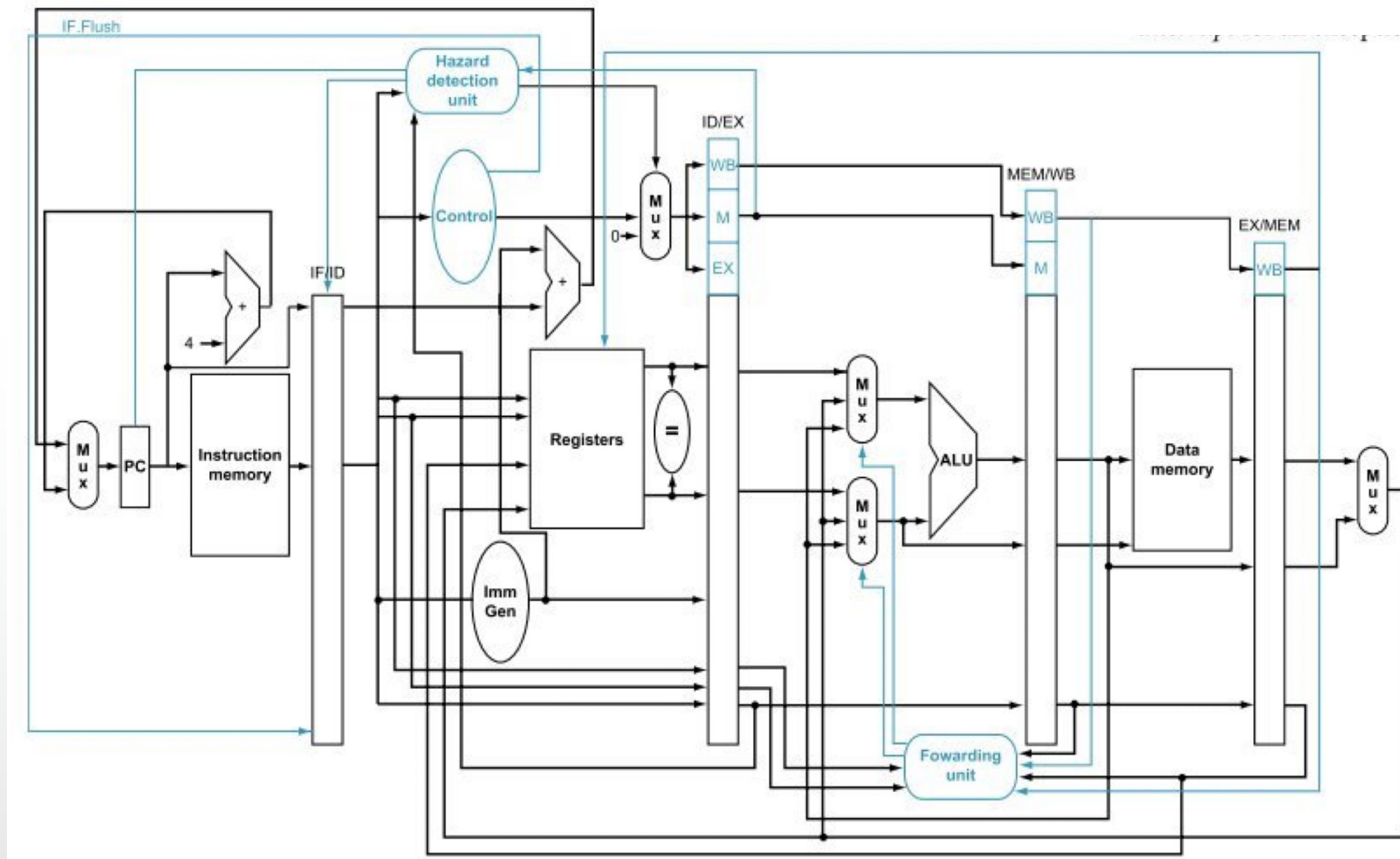
Forwarding Unit

- Example. ForwardB controls the lower MUX
- Pseudo code

```
if (EX/MEM.RegWrite and  
    (EX/MEM.RegisterRd != 0) and  
    (EX/MEM.RegisterRd == ID/EX.RegisterRs2))  
    ForwardB = 10
```

- Selects ALU operand from the prior ALU result
- Similarly for MEM hazard
- Can easily be coded in combinational SystemVerilog

Hazard Detection



Detection Logic

- Pseudo code

```
if (ID/EX.MemRead and  
    (ID/EX.RegisterRd == ID/EX.RegisterRs1) or  
    (ID/EX.RegisterRd == ID/EX.RegisterRs2))  
    stall the pipeline
```

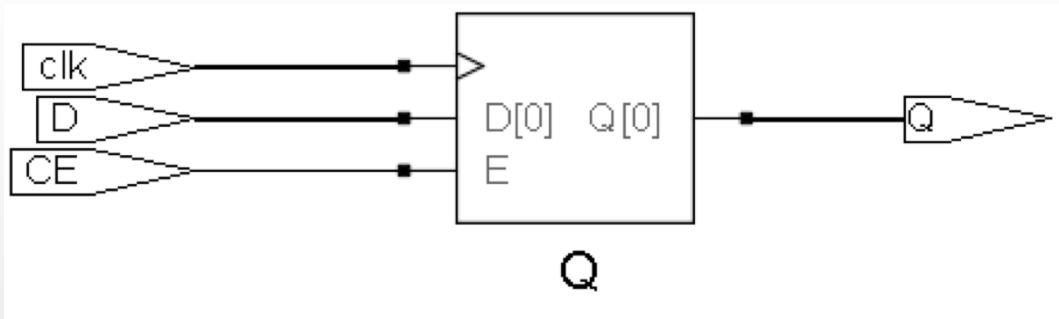
- Stalling the pipeline means preventing the PC and IF/ID registers from changing

Flip-flop with enable

```
// rising edge triggered D-type flip-flop with clock enable
module ff4 (input logic clk, D, CE, output logic Q);

always_ff @ (posedge clk)
    if (CE)
        Q <= D;

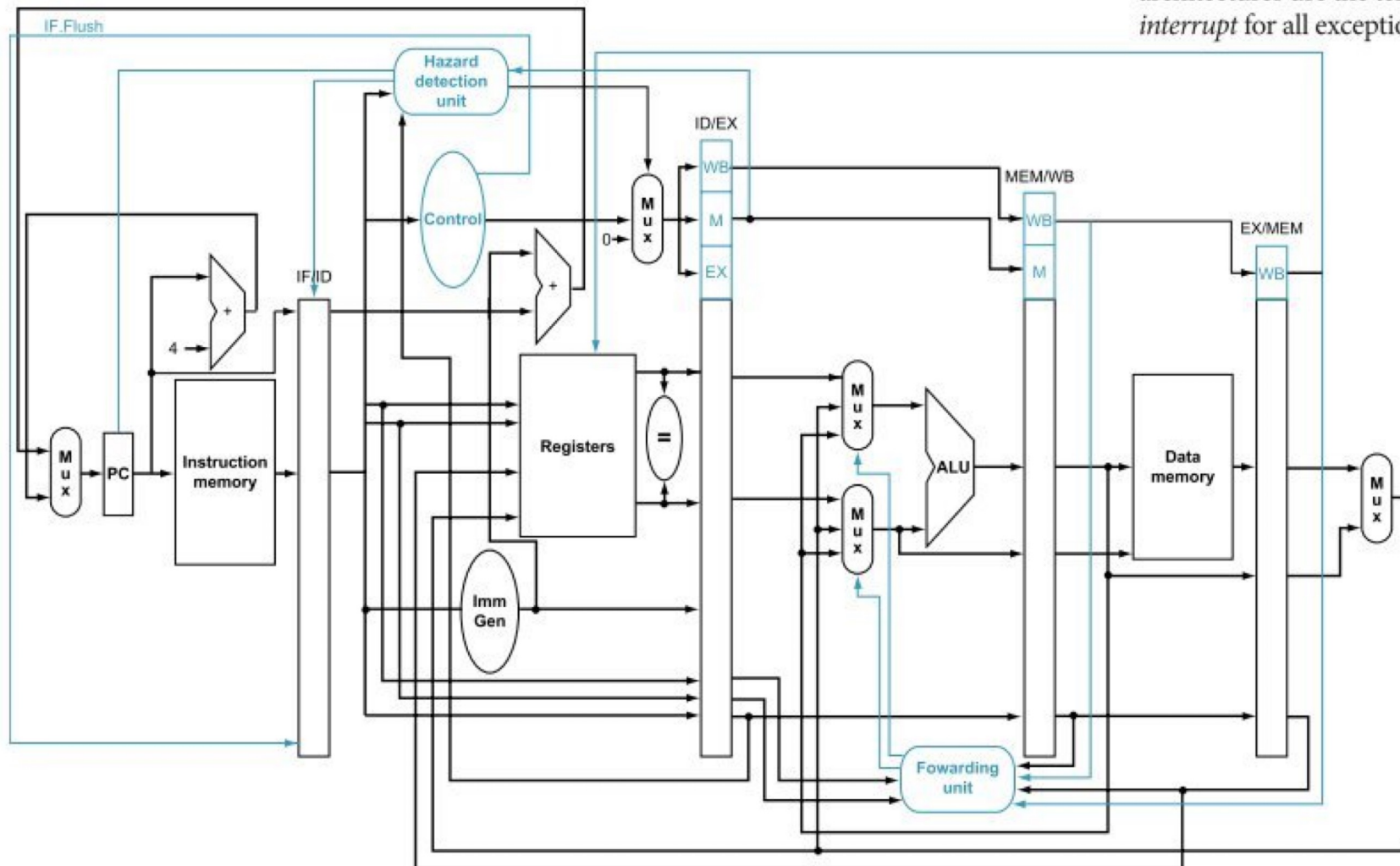
endmodule
```



If CE is false, the contents of the flip-flop (or register) don't change

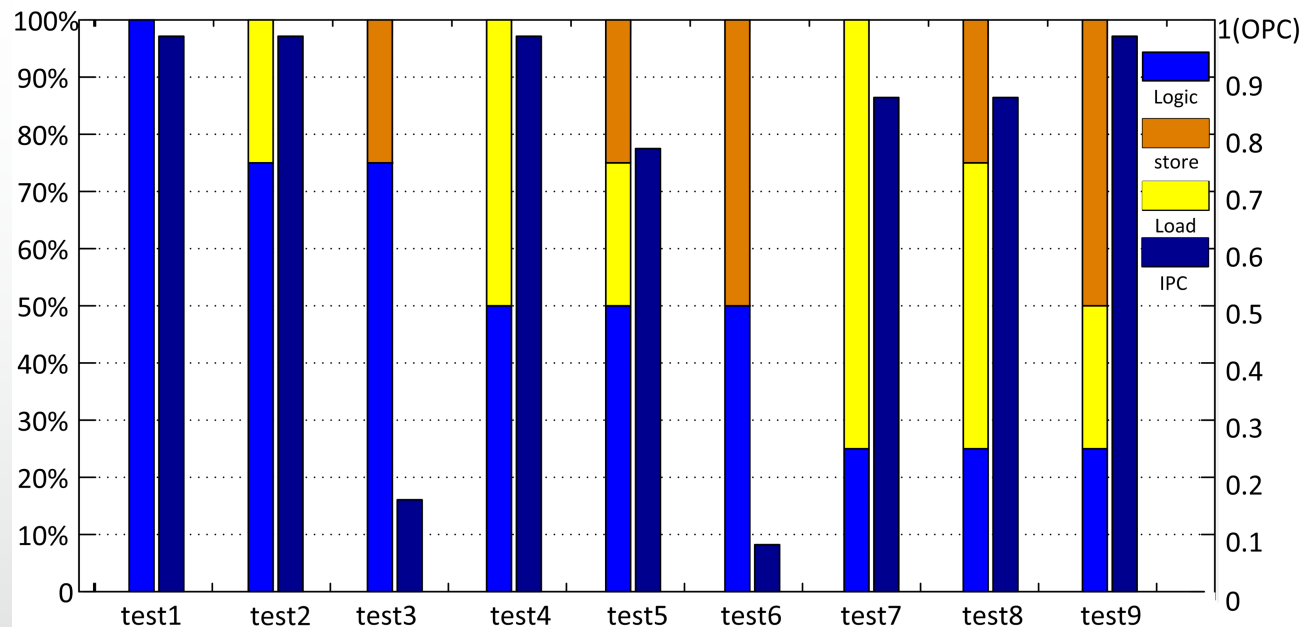
Complete CPU

architectures use the term
interrupt for all exceptions.



Power and Energy

- For a pipelined RISC processor, the energy for each instruction is approximately equal, except for *STORE*.
 - STORE* means writing data off chip, more energy
 - STORE* is slow, so the *power* may be less



IPC – Instructions per cycle
See test 6 – a lot of STORE instructions – low IPC